MIDS ASSIGNMENT

FUNDAMENTAL PROGRAMMING

STUDENT NAME

HaFiz Zohaib Memon

DEPARTMENT

BS ARTIFICIAL INTELLIGENCE (AFTERNOON)

ROLL NUMBER

115-6323

PROGRAMMING TASK: USER-DEFINED MARKSHEET GENERATOR

QUESTION: Build a C++ program that creates a marksheet generator. The program should: 1) Accept student name, roll number, and marks for 5 subjects (Math, Physics, English, Chemistry, Biology) 2) Calculate total marks and percentage 3) Assign grades based on percentage 4) Display a formatted marksheet output

PROGRAM REQUIREMENTS

- Take input for student name and roll number
- Accept marks for 5 subjects
- Store subject marks using variables or arrays
- Compute total obtained marks
- Calculate percentage from the total
- Assign grade according to percentage
- Print a neat and readable marksheet

KEY CONCEPTS

- Input and output handling in C++
- Variables, data types, and formatting
- Arithmetic operations and formulas
- Conditional statements for grade assignment
- Basic program structure with main()
- User-defined logic for report generation

EXPECTED OUTPUT

Student Name: HaFiz Zohaib Memon

Roll Number: 115-6323

Subjects: Math, Physics, English, Chemistry, Biology
Total Marks: Calculated from all 5 subjects
Percentage: Displayed as a whole or decimal value
Grade: Assigned using the final percentage

ANSWER: The C++ program implements a marksheet generator with the following features:

Welcome | Student_marksheet.cpp

Student_marksheet.cpp

```cpp
#include <iostream>
#include <string>
#include <iomanip>

using namespace std;

void Marksheet() {
    string studentName;
    int studentId;
    float Math, Physics, English, Chemistry, Biology;
    float totalObtained, totalPercentage;
    string Grade;

    cout << "-----------------------------------" << endl;
    cout << "       STUDENT DATA ENTRY          " << endl;
    cout << "-----------------------------------" << endl;

    cout << "Enter Your Name: ";
    cin >> studentName;
    for (int i = 0; i < studentName.length(); i++){
    if (studentName[i] >= '0' && studentName[i] <= '9'){
        cout <<"Invalid Name! Only Characters Allowed.\n";
        cout <<"Enter Your Name Again: ";
        cin >> studentName;
        i = -1;
    }
}

    cout << "Enter Your Roll No: ";
    cin >> studentId;
    while (cin.fail() || studentId <= 0){
        cin.clear();
```

Welcome | Student_marksheet.cpp

Student_marksheet.cpp > ...

```cpp
void Marksheet() {
    else if (totalPercentage >= 80) Grade = "A";
    else if (totalPercentage >= 70) Grade = "B";
    else if (totalPercentage >= 60) Grade = "C";
    else if (totalPercentage >= 50) Grade = "D";
    else Grade = "Fail";

    cout << "\n\n\tBOARD OF INTERMEDIATE & SECONDARY EDUCATION HYDERABAD" << endl;
    cout << "\t\t\t MARKS CERTIFICATE" << endl;
    cout << "-----------------------------------------------------------------" << endl;
    cout << "NAME: " << studentName << "\t\t\t\tSEAT NO: " << studentId << endl;
    cout << "-----------------------------------------------------------------" << endl;
    cout << "SUBJECTS\t\tMAX MARKS\tMIN MARKS\tOBTAINED" << endl;
    cout << "-----------------------------------------------------------------" << endl;
    cout << "ENGLISH\t\t\t100\t\t33\t\t" << English << endl;
    cout << "MATHEMATICS\t\t100\t\t33\t\t" << Math << endl;
    cout << "PHYSICS\t\t\t100\t\t33\t\t" << Physics << endl;
    cout << "CHEMISTRY\t\t100\t\t33\t\t" << Chemistry << endl;
    cout << "BIOLOGY\t\t\t100\t\t33\t\t" << Biology << endl;
    cout << "-----------------------------------------------------------------" << endl;
    cout << "TOTAL\t\t\t500\t\t\t\t" << totalObtained << endl;
    cout << "-----------------------------------------------------------------" << endl;

    cout << "RESULT: " << (totalPercentage >= 33 ? "PASS" : "FAIL") << endl;
    cout << "PERCENTAGE: " << fixed << setprecision(2) << totalPercentage << "%" << endl;
    cout << "GRADE: " << Grade << endl;
    cout << "-----------------------------------------------------------------" << endl;
}

int main() {
    Marksheet();
```

Activate

Algorithm Explanation

Theprogram begins by acceptingthestudents' name, roll number, and marks for five subjects: Math, Physics, English, Chemistry, and Biology. It then storesthe marks,adds them to compute the total, and divides the obtained marks by the maximum possible marks to calculate the percentage. Finally, it displays a formatted marksheet containing the student details, total marks, percentage, and grade.

The logic is straightforward and follows a clear sequence: input, processing, and output. This makes the program easy to understand, maintain, and expand ifadditional subjects orgrading rules are required in the future.

Output Format

The output displays the student information, subject-wise marks, total marks, percentage, and final grade in a clear marksheet format. This layout includes a professional header at the top that shows the student's name and roll number. Below that, each subject is listed in a simple table with its specific marks to make it easy to read. The system automatically calculates the total score and the overall percentage, then assigns a final grade like A, B, or C based on the results. Everything is organized in a clean and minimal design so the final result is clear at a single glance

```
      STUDENT DATA ENTRY
-----------------------------------
Enter Your Name: zohaib
Enter Your Roll No: 115
Enter Math Marks: 90
Enter Physics Marks: 95
Enter English Marks: 85
Enter Chemistry Marks: 70
Enter Biology Marks: 80


        BOARD OF INTERMEDIATE & SECONDARY EDUCATION HYDERABAD
                          MARKS CERTIFICATE
-----------------------------------------------------------------
NAME: zohaib                            SEAT NO: 115
-----------------------------------------------------------------
SUBJECTS                MAX MARKS       MIN MARKS       OBTAINED
-----------------------------------------------------------------
ENGLISH                 100             33              85
MATHEMATICS             100             33              90
PHYSICS                 100             33              95
CHEMISTRY               100             33              70
BIOLOGY                 100             33              80
-----------------------------------------------------------------
TOTAL                   500                             420
-----------------------------------------------------------------
RESULT: PASS
PERCENTAGE: 84.00%
GRADE: A
```